EE-102 Electric Circuit Analysis
Experiment # 10

Implementation of Thevenin Theorem

Objective/s:

- To verify Thevenin theorem

Components

- DC power supply
- Ohmmeter
- DC ammeter
- DC voltmeter
- Resistances of values 47Ω, 220Ω, 330Ω, 1KΩ

1. Theory

Thevenin's Theorem states that **it is possible to simplify any linear circuit, no matter how complex, to an equivalent circuit with just a single voltage source and series resistance connected to a load.** Thevenin's Theorem is especially useful in analyzing power systems and other circuits where one resistor in the circuit (called the "load" resistor) is subject to change, and re-calculation of the circuit is necessary with each trial value of load resistance, to determine voltage across it and current through it.

Thevenin's Theorem makes this easy by temporarily removing the load resistance from the original circuit and reducing what's left to an equivalent circuit composed of a single voltage source and series resistance. The load resistance can then be re-connected to this "Thevenin equivalent circuit" and calculations carried out as if the whole network were nothing but a simple series circuit.

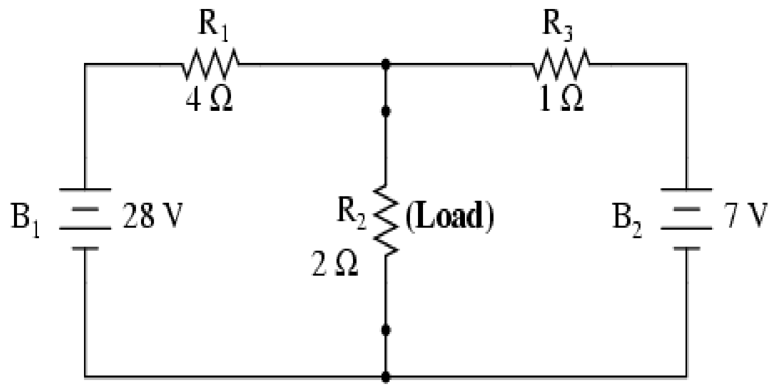


Figure 1

. . . after Thevenin conversion . . .

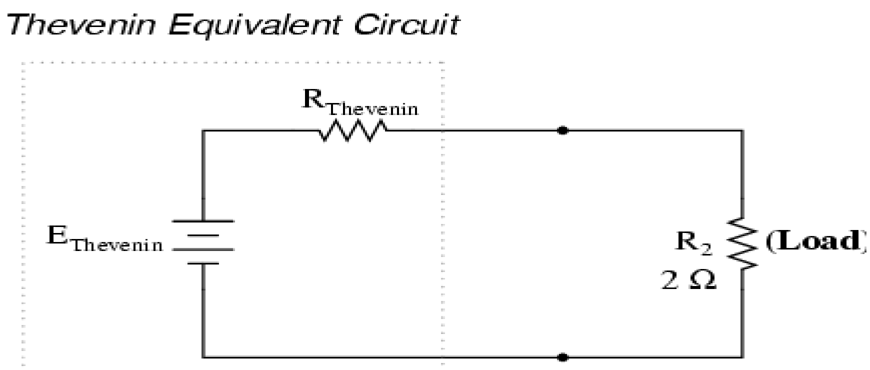


Figure 2

The "Thevenin Equivalent Circuit" is the electrical equivalent of B_1, R_1, R_3, and B_2 as seen from the two points where our load resistor (R_2) connects.

The Thevenin equivalent circuit, if correctly derived, will behave the same as the original circuit formed by B1, R_1, R_3, and B_2. In other words, the load resistor (R_2) voltage and current should be the same for the same value of load resistance in the two circuits. The load resistor R_2 cannot "tell the difference" between the original network of B_1, R_1, R_3, and B_2, and the Thevenin equivalent circuit of $E_{thevenin}$, and $T_{hevenin}$, provided that the values for $E_{thevenin}$ and $R_{Thevenin}$ have been calculated correctly.

2. Procedure

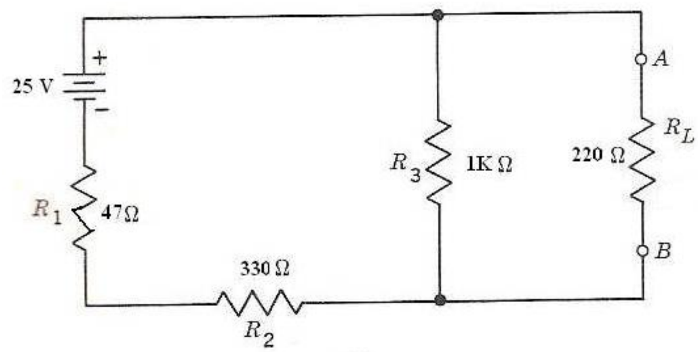
1. Connect the circuit as shown in the above fig 1
2. Remove the R_L from the circuit across which the Thevenin equivalents have to be evaluated.
3. To find the R_{TH} remove the voltage-source from the circuit and short circuit the terminals from which the supply was connected. Now place an ohmmeter across the terminals A and B.

4. Also confirm the value of the equivalent resistance from calculations and record it in the table.

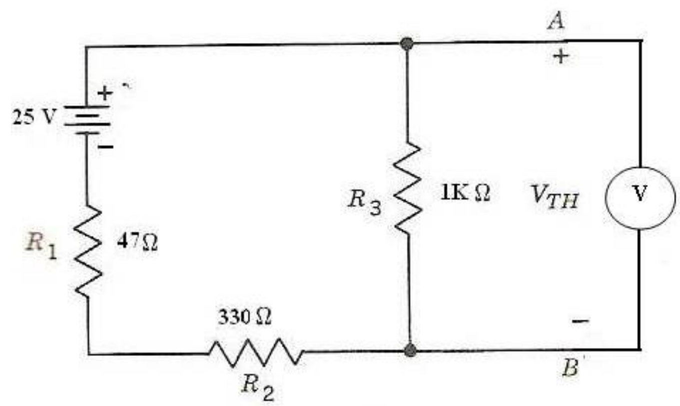
5. To find the value of V_{TH} place a voltmeter across the terminals A and B after retaining the voltage source in the circuit back.

6. Also confirm the value of the equivalent voltage from calculations and record it in the table.

7. Now construct the Thevenin equivalent circuit and measure VAB and IL. Also calculate these values by using methods other than Thevenin theorem.



(a)



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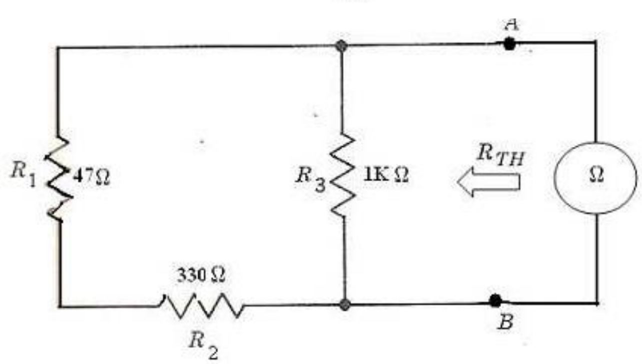

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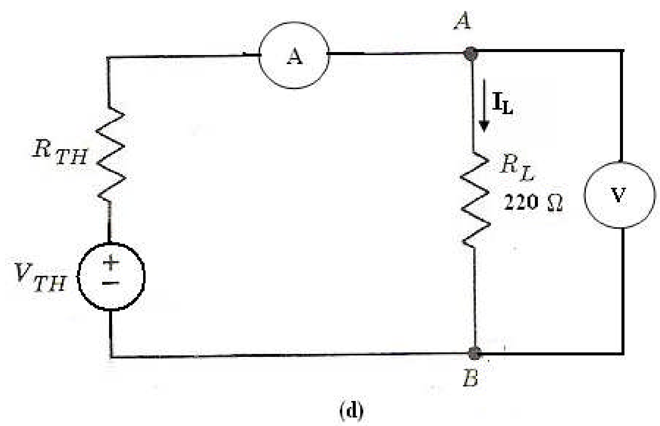

Figure 3

3. Observations

1.1. For Resistor:

S.No.	Nominal Value (Ω)	Measured Value (Ω)
1.	47	
2.	220	
3.	330	
4.	1000	

Table 1

1.2. For Thevenin theorem

R_L	V_{TH} (Volts)		R_{TH} (Ohms)		I_L (A)		V_{AB} (V)	
	Measured	Commuted	Measured	Commuted	Measured	Commuted	Measured	Commuted

Table 2